

Exercise 7: Financial Forecasting

Recursion is a technique in which a method calls itself repeatedly until a base condition is reached. It helps solve problems by breaking them into smaller and simpler subproblems. Recursive solutions are often easier to understand and implement for problems involving repeated calculations.

2. Implementation:

```
public double futureVal(double value, double rate, int years){  
    if(years == 0){  
        return value;  
    }  
    return futureVal(value * (1 + rate), rate, years - 1);  
}
```

Analysis:

The recursive method calculates the future value year by year until the required number of years is reached.

- Best Case: $O(1)$ when years = 0
 - Average Case: $O(n)$
 - Worst Case: $O(n)$
- where n represents the number of years.

Optimization

The recursive solution creates a new function call for each year, which increases memory usage due to the call stack. For a very large number of years, an iterative approach can be used to avoid excessive recursion. Another optimization is to use mathematical formulas or memoization techniques to reduce unnecessary computations and improve performance.